

Teaching computers to check code safely

A small, working demo that combines two open security standards — and shows how to make an AI **prove its work** before anyone trusts it.

 **Like a building inspection:** the **building code** is the rulebook of what's safe; a **trustworthy inspector** never guesses, always shows evidence, gets a second opinion, and leaves the owner in charge; **hello-spec** is the model house we walk through to see it work.

THREE PIECES FIT TOGETHER



PROJECT CODEGUARD

The building code

A shared rulebook of safe-coding practices. Knowledge, written down.



FOUNDRY SECURITY SPEC

The trustworthy inspector

A blueprint for an AI that only reports problems it can **prove**.



HELLO-SPEC

The model house

A tiny demo that runs both on a flawed program — and a safe twin.

THE LIFE OF A FINDING — EVERY PROBLEM MUST SURVIVE THESE STEPS

1

Spot

Notices something risky

2

Prove

Points to the evidence

3

Confirm

A 2nd checker agrees

4

Write up

Only proven ones

5

Propose

A tested fix

6

You decide

Nothing auto-changes

● From CodeGuard — the rules

- Safe-coding rules in CodeGuard's own format
- Checked & packaged by CodeGuard's real tools
- The fixes apply CodeGuard's safe practices

● From Foundry — the inspector

- A team of specialist inspectors
- "Prove it" before anything is called real
- An independent second inspector confirms
- 11 trust rules it must always obey



Every finding proven · Independent re-check · Human in control

The cleverness of AI, harnessed — and only believed when it can show the proof.

A learning project combining Project CodeGuard + the Foundry Security Spec.

The inspected program is a teaching example with deliberate flaws — not a real product.